

Given two vectors u and v in $\mathbb{R}^2$, their sum is the vector $u+v$ obtained by adding corresponding entries of u and v . For example,

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1+2 \\ -2+5 \end{bmatrix}$$

Given a vector u and a real number c , the scalar multiple of u by c is the vector cu obtained by multiplying each entry in u by c . For instance, if $u = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ and $c=5$, then $cu = 5 \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 15 \\ -5 \end{bmatrix}$

Example 1: Given $u = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $v = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$, find $4u$, $(-3)v$ and $4u + (-3)v$

Solution: $4u = \begin{bmatrix} 4 \\ -8 \end{bmatrix}$ $(-3)v = \begin{bmatrix} -6 \\ 15 \end{bmatrix}$

$$4u + (-3)v = \begin{bmatrix} 4 \\ -8 \end{bmatrix} + \begin{bmatrix} -6 \\ 15 \end{bmatrix} = \begin{bmatrix} -2 \\ 7 \end{bmatrix}$$

Geometric Descriptions of $\mathbb{R}^2$

Consider a rectangular coordinate system in the plane. Because each point in the plane is determined by an ordered pair of numbers, we can identify a geometric point (a,b) with the column vector $\begin{bmatrix} a \\ b \end{bmatrix}$.

So we may regard $\mathbb{R}^2$ as the set of all points in the plane.

See Figure 1.

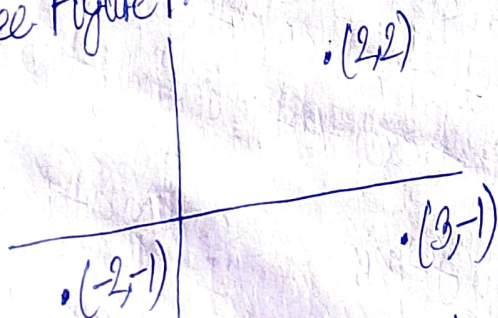


Figure 1: Vector as points

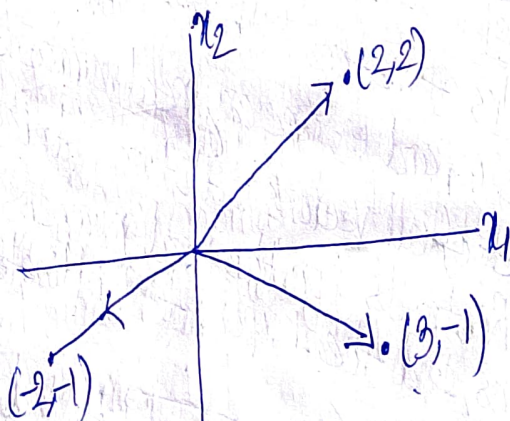


Figure 2: Vector with arrows